

TECHNICAL REPORT

Coverage-hole waivers at COREV_PULP=0 (the TRL5 scope)

openhwgroup/cv32e40p

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Prepared by	Rivoryxa Technologies
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Website	www.rivoryxatechnologies.com

1 Answer

We rechecked the 15 open coverage-hole issues with COREV_PULP=0, the configuration that stays in scope for TRL5 once the PULP and XCV extensions move to TRL4.

Outcome at COREV_PULP=0	Count	Issues
The hole disappears. The code is inside if (COREV_PULP) or tied to 0.	4	#1005, #1006, #1009, #1010 (all hardware-loop logic)
Still present, still proven unreachable. The waiver carries over.	10	#1007, #1011, #1012, #1015, #1016, #1017, #1018, #1019, #1022, #1023
Present and reachable. It needs a test, not a waiver.	1	#1004 (a directed test is supplied)

Three of the ten carry a condition. Read the exceptions below before reusing their waiver text.

2 The question

At the CVE4 meeting (September 2026), the group agreed to move the PULP extensions (configurations P_F0, P_Z0) and the XCV instructions to TRL4. The TRL5 configurations therefore build the core with COREV_PULP=0.

The open coverage-hole issues were filed against a COREV_PULP=1 build. The question is whether each hole and its waiver still holds in the TRL5 build.

3 What we did

For each of the 15 issues, we started from the existing disposition, rebuilt the design with COREV_PULP=0 and reran the same checks:

- Structural check. Where the code sits inside a COREV_PULP generate or if, we matched the source lines that remove it. This is a reading of the RTL, not a proof.
- Formal proof. For each remaining hole, we reran the unreachable assertion as an unbounded proof (SymbiYosys, abc pdr unless noted), together with a cover. The cover shows that the surrounding state is reachable, so the proof is not true merely because nothing can happen.
- Reachability check. Where the hole was already known to be reachable, we reran the bounded model check and the directed simulation test.

A prediction for each hole was written before any run. All 15 held.

4 Results

Issue	What the hole is	Site (6033d2b)	At COREV_PULP=0	How it was checked	Earlier report
#1004	Interrupt during the first fetch after wake-up	controller.sv:399	Reachable (one of its two conditions)	irq && debug_mode_q: proof PASS. irq && debug_req_pending; BMC counterexample at depth 30, plus a directed test hit	PDF
#1005	Outer loop ending inside the inner loop	controller.sv:640	Disappears	gen_no_hwlp ties hwlp_end*_eq_pc to 0 (controller.sv:1306-1311)	PDF
#1006	Duplicate hardware-loop jump guard	controller.sv:632, 642	Disappears	Same tie-off as #1005	PDF
#1007	Single-step with a stalled decode stage	controller.sv:675	Still unreachable	p_line675 proof PASS	PDF
#1009	Nested hardware-loop end addresses too close together	controller.sv:831	Disappears	Line 831 is in state DECODE_HWLOOP, whose whole body is inside if (COREV_PULP) (lines 717-899)	PDF
#1010	Debug single-step inside a hardware loop	controller.sv:850	Disappears	Same DECODE_HWLOOP body as #1009	PDF
#1011	Debug entry with no recorded cause	controller.sv:1187, 1210	Still unreachable	Both assertions proof PASS; covers reached	PDF
#1012	Debug flush with an impossible single cause	controller.sv:1241	Still unreachable, with a condition	Rows 5 and 10: abc pdr PASS. Row 4: k-induction (k=12, smtbmc yices) PASS under an assumption	PDF
#1015	Operand-select combination in the ID stage	id_stage.sv:872	Still unreachable	Proof PASS at FPU=0 and at FPU=1	PDF
#1016	Write-back contention flag in the EX stage	ex_stage.sv:211	Still unreachable	Proof PASS; antecedent cover reached	PDF
#1017	Result memorisation enable in the EX stage	ex_stage.sv:387	Still unreachable	Proof PASS; antecedent cover reached	PDF
#1018	Result memorisation clear in the EX stage	ex_stage.sv:396	Still unreachable	Proof PASS; antecedent cover reached	PDF
#1019	Ready handshake in the divide and square-root unit	fpnew_divsqrt_th_32.sv:288	Still unreachable	Proof PASS; antecedent cover reached	PDF
#1022	Arbitration between operation groups in the FPU	rr_arb_tree.sv (gen_fair_arb, lzc)	Cause still holds, with a condition	Proof PASS (the arbiter never has more than one requester)	PDF
#1023	Two-cycle APU write-back path in the EX stage	ex_stage.sv:237, 241	Still unreachable, with a condition	Proof PASS at FPU latency 0	PDF

All ten "still unreachable" results are unbounded proofs, not bounded checks. No run timed out and no tool reported an error.

Exceptions: where "the waiver carries over" needs a qualifier

Issue	What to know
#1012 , row 4	Proven by k-induction under an assumption that the trigger-match input is stable, the same assumption as at COREV_PULP=1. The waiver is valid only if that assumption is accepted. A plain abc pdr proof did not finish in over 30 minutes, because the cone includes the 32-bit trigger comparator.
#1022	The cause is proven, but there is no line-exact waiver to carry over. The issue names no module instance or line, so mapping the proof onto specific uncovered lines needs the original coverage database.
#1023	Proven for FPU latency 0 only. It is not established for the pipelined FPU configurations (latency 1 or 2).

Which FPU configurations these results cover

Holes	Built with
Controller and ID stage (#1004 to #1015)	FPU=0. #1015 was also proven at FPU=1.
FPU and EX-stage (#1016 to #1023)	FPU=1, FPU latency 0
Any hole at FPU latency 1 or 2	Not run

#1004: the one hole that is reachable

The hole has two conditions. `irq_req_ctrl_i` && `debug_mode_q` is proven unreachable. `irq_req_ctrl_i` && `debug_req_pending` is reachable at `COREV_PULP=0`: the core wakes from wfi on an interrupt while a debug request is pending, and lands in `FIRST_FETCH` with both set. The evidence:

Evidence	Result
Bounded model check, <code>COREV_PULP=0</code>	Counterexample at depth 30
Directed test, core-v-verif core Verilator testbench, <code>COREV_PULP=0</code>	1 hit; program EXIT SUCCESS (32 interrupts and 32 debug entries taken)
Same counter over the 94 ACT4 programs (cv32e40p-dv-review testbench, I0)	0 hits
Same counter with the directed test added	1 hit

The 0 over ACT4 says nothing about the RTL. That testbench ties `debug_req_i` to 0, so ACT4 never sends a debug request. Diagnostic counters on the debug-request path also read 0 there. The directed test reaches the core only with the wiring patch from [report 04](#). The test and its files are in [tests/cv32e40p-hole-1004](#).

#1010: where it stands upstream

Rows of line 850	Status
Rows 2 and 3 (single-step on, debug mode off)	Proven unreachable. Assertion submitted as PR #1073.
Row 4 (single-step on, in debug mode)	Reachable. Closed by the directed test tests/cv32e40p-1010-row4 : 28 cycles on row 4, 0 in the control run.
Formal bind file	PR #1072 fixes the clock and reset names in the controller bind, so the upstream formal setup can check the controller.

Both pull requests target dev, were opened on 24 September 2026, and have no review yet. At `COREV_PULP=0` the whole of line 850 disappears, because it lives in the hardware-loop state.

5 What this means for TRL5

- The four hardware-loop holes leave the TRL5 scope. No waiver is needed for them at `COREV_PULP=0`. Their waivers matter only for the TRL4 PULP configurations.
- Ten waivers carry over to the TRL5 build. Seven need no qualifier. #1012 row 4, #1022 and #1023 need the qualifiers above.
- #1004 needs a test in the regression, not a waiver. The directed test hits it. Whether the core behaves correctly when it gets there is a separate question that this check does not answer. The program completes normally and takes every interrupt and debug request.
- The pipelined FPU configurations are not covered. If the TRL5 set includes FPU latency 1 or 2, then #1023 and the FPU-side holes (#1016 to #1022) need a rerun at those latencies.

6 How to repeat

Item	Where
#1004 checker, environment constraints, directed test and probe	tests/cv32e40p-hole-1004 , whose README gives the bind lines and the SymbiYosys settings
#1010 row 4 directed test	tests/cv32e40p-1010-row4
Checkers and proofs for the other holes	Described in each issue's PDF in reports/ (property text, engine, depth, cover)

Our harness that drives all 15 is not public. To rerun one proof without it:

1. Read the RTL with the yosys-slang front end (or convert it with sv2v), with COREV_PULP=0 and the FPU setting from the table above.
2. Bind the checker and the environment constraints.
3. Run SymbiYosys:
 - for an unreachable proof, mode prove, engine abc pdr;
 - for #1012 row 4, k-induction with smtbmc yices at depth 12;
 - for #1004 reachability, mode bmc at depth 30.
4. Prepare the design with prep; flatten; memory_map; async2sync; techmap; setundef -anyseq.

Tools: Yosys 0.69, SBY 0.69, sv2v, yosys-slang, Verilator 5.050.

7 Limits

- Structural "disappears" verdicts are a source reading, not a proof. The four hardware-loop holes rest on matching the generate and if (COREV_PULP) lines.
- The clock gate is modelled as transparent. That is sound for unreachable, not for timing questions.
- The proofs hold under the environment constraints in env_constraints.sv. These are the OBI handshake rules, at most two outstanding transactions, no scan, and fetch enabled. They mirror the upstream formal assumes, but a waiver reviewer should confirm they match the integration.
- The formal trace for the #1004 counterexample at COREV_PULP=0 was not kept. The summary result was kept, and the simulation hit reproduces it.
- This checks the 15 known holes only. Coverage of the COREV_PULP=0 build as a whole was not measured here.
- Nothing here has been reviewed or accepted by OpenHW. It is an investigation record, not a sign-off.

8 Files

- This report: cv32e40p-trl5/03-coverage-holes-trl5-scope.md
- tests/cv32e40p-hole-1004/: checker_1004.sv, env_constraints.sv, probe_hole1004.sv, first_fetch_dt.c, dbg.S, README.md
- tests/cv32e40p-1010-row4/: directed test for #1010 row 4
- Per-issue reports: reports/